LAB 08 – Wireless Exploitation

Nicholas Kehayias & Dustin Crawford

IT203

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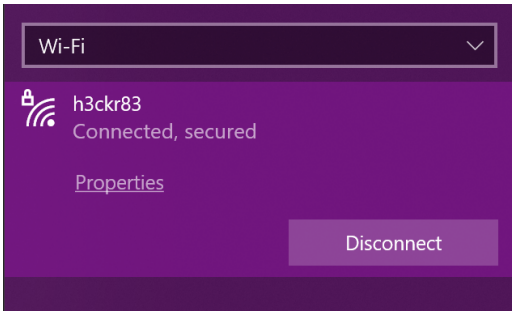
Introduction:

In this lab, we will be attempting to crack wireless network security measures in an effort to gain access to local networks (in our case the Asus router provided). We prepared for the tasks required for the lab by ensuring we have the proper virtual machine setup as well as the correct WiFi adapter settings.

Exercise 1

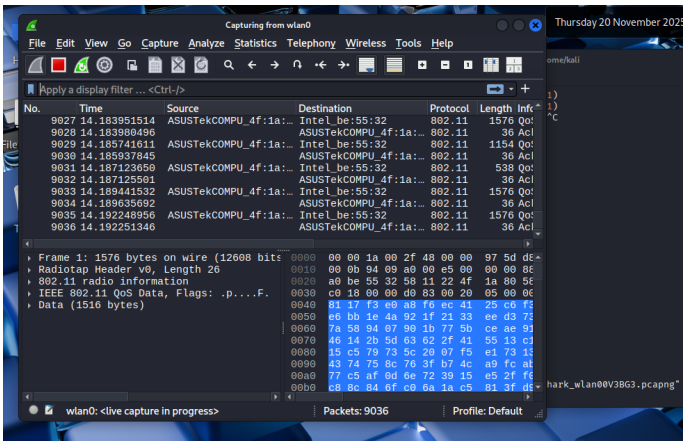
Task 1: Setting up the Wireless Router with a laptop

Image of the WiFi Network



Task 2: Setting up the Wireless Adapter with Kali Linux

Image of traffic captured on Monitor Mode in Wireshark



Answer the following questions:

1. What is the difference between managed mode and monitor mode in wireless adapters? Managed mode is for normal wireless connectivity, like when you're browsing on social media or need normal internet access. Monitor mode is for capturing traffic on a network for observation without joining the network.
2. What is WiFite and what is it used for? WiFite is an open source tool used to test the security of wifi networks.

Key Learnings:

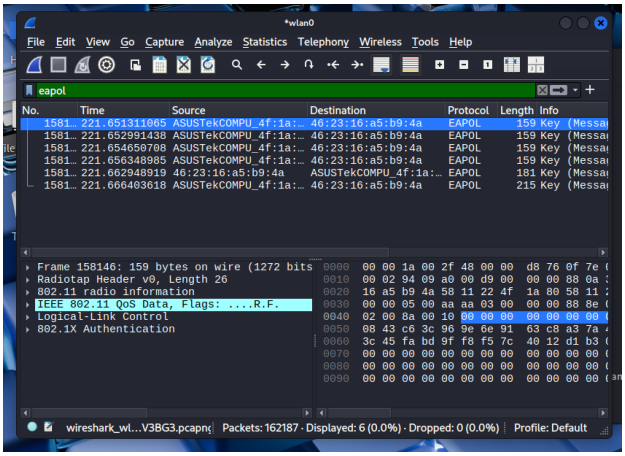
- **How to set up wifi network**

- **How to use network adapter in Linux/Host PC sharing**
- **How to enable monitor mode and watch traffic**

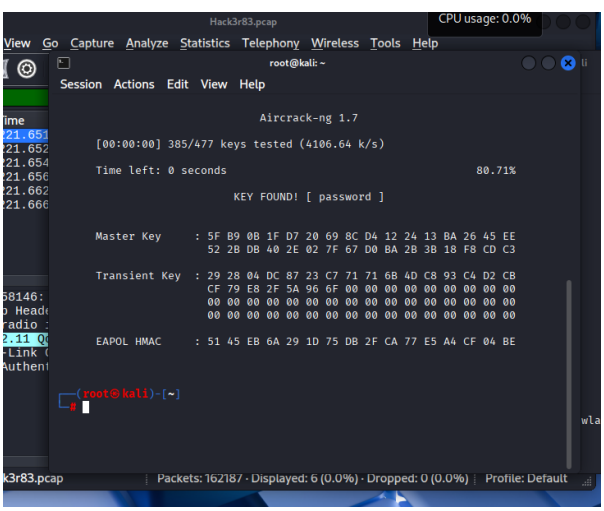
Exercise 2

Task 1: Capturing Four-Way Handshake (Authentication Packages)

Captured EAPOL handshake in Wireshark



Password ‘password’ cracked using Aircrack



Answer the following questions:

1. Explain the EAPOL packet type? When is it used? Why did we need to filter for it in monitor mode? EAPOL is a network protocol for secure network access, primarily used in Wi-Fi security like WPA/WPA2. It plays a key role during the 4-way handshake process to exchange the encryption keys needed for a secure connection. We filtered for EAPOL packets in monitor mode to isolate critical authentication messages

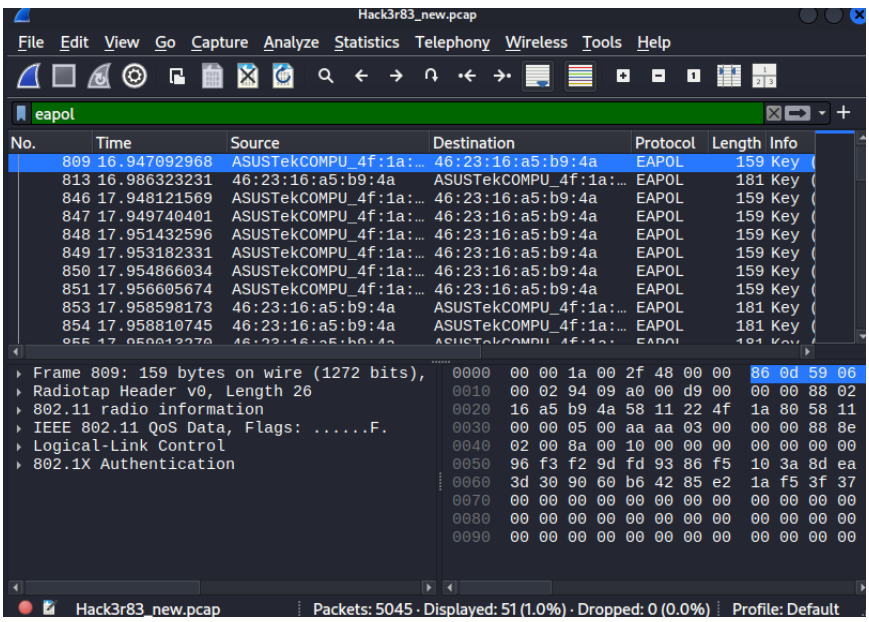
2. What is Aircrack-ng tool? Other than cracking WPA passwords, what can it do?
   Aircrack-ng tool is a suit of command line tools. It provides capabilities for comprehensive wireless network analysis and security assessment, from monitoring airwaves to actively manipulating network traffic.

Key learnings:

- **How to capture EAPOL authentication traffic using wireshark**
- **How to crack the password using Aircrack**

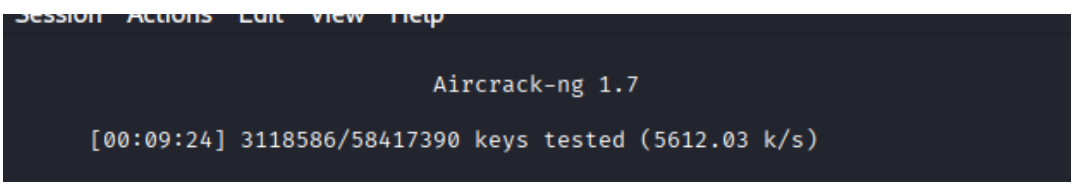
Exercise 3

Task 1: Repeating the Attack with a Stronger Password



Task 2: Finding a Better Wordlist

Many more keys to try after downloading and compiling the Github list.



Task3: Cracking the New Password

Cracked the new password using Wifite

```
      Time left: 0 seconds
80.00%
                          KEY FOUND! [ eAPBXCIw71 ]


      Master Key     : 53 AA A9 A9 14 00 F1 5D 1D 28 A9 D7 4C 96
 02 30
C4 40 49 F9 C2 74 24 D6 08 88 3E B0 2E E3 A2 CC
      Transient Key  : AC 89 7F 5C D4 5F 32 4D A7 F5 06 91 14 3F
 9A 5F
                       94 D5 E6 8C F6 5E F9 E7 5F 06 A8 BF 41 37
 70 9B
                       91 29 E5 31 FE 27 EE 50 8A 98 BE 5B 12 A6
 FA 8AEAPOL HMAC     : C5 36 A7 39 12 E0 07 F5 60 1F A0 9E D7 AD
 ED 4C                 F9 C2 8C 31 4C F8 24 7D 87 B7 91 AF 72 95
 B0 BE
```

Answer the following questions:

1. Explain Each part of the screen below?

   The top line indicates the status of how many keys have been tried over how many total there are.

   Time left is a prediction of how long it will take for Aircrack to make its way through all the keys.

   Key found is the wifi password successfully cracked

   Master key is used for WPA2 encryption.

   Transient key is used to encrypt the individual packets during the 4-way handshake.

   When the EAPOL HMAC hash matches, the password is confirmed as correct.

2. Explain the changes we made to find the new password?

   They help us find the new password because it is a significantly larger dictionary of keys to try

3. How does the new text file (**`~/wpa2wordlists/Wordlists/Crackdown2016/full.txt`**) help us?

   The text file is what Aircrack uses to test to test thousands of passwords.

   Aircrack is not cracking on its own, rather it is testing each password and verifying if the hash matches.

4. What does hashing a password mean?

Turning a plaintext password into an encrypted string of characters.

5. What are rainbow tables?

Store the results of hash functions to quickly decrypt passwords.

6. What is HMAC? What does the HMAC value below represent?

Hash-based message authentication code - combines a hash function with a secret key to keep a messages integrity

7. What does Aircrack-ng actually do to find the password?

It captures data packets to perform cryptanalysis on the data it gathers.

Key Takeaways:

- **How to create a large file with all contents of other files**
- **How to once again use Aircrack to crack a harder password**
- **What exactly Aircrack is doing and HOW it does it**

Conclusion:

In this lab, we successfully captured a WPA2 handshake and used aircrack-ng to recover the wireless network password. After fixing the wordlist path and providing a more complete dictionary file, aircrack-ng was able to match the stronger password against the captured EAPOL handshake and reveal the correct key. This process demonstrated how WPA2 password cracking relies entirely on the quality and correctness of the wordlist used.